



Lecture 10: Conditional expectation and transforms

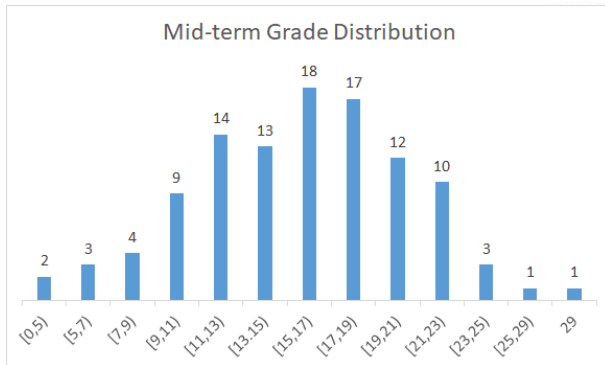
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- Mean: 15.49
- Standard deviation: 4.81
- If you want to view your own answer sheet, please contact TAs in advance and come to 8-208 or 11-201, Rohm Building this Wednesday or Thursday after the recitation session.



1. Review
2. Conditional expectations
3. Transforms
4. Recitation
5. Summary



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- Convolution

$$p_W(w) = \sum_x p_X(x)p_Y(w-x)$$

$$f_W(w) = \int_{-\infty}^{\infty} f_X(x)f_Y(w-x)dx$$

- Covariance

$$\text{cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}[X]) \cdot (Y - \mathbf{E}[Y])]$$

- Correlation

$$\rho = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y} \in [-1, 1]$$

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Definition of conditional expectations

Consider the RV $\mathbf{E}[X|Y]$, which takes the value

$$\mathbf{E}[X|Y = y] = \begin{cases} \sum_x xp_{X|Y}(x|y), & X \text{ discrete,} \\ \int_{-\infty}^{\infty} xf_{X|Y}(x|y) dx, & X \text{ continuous.} \end{cases}$$

when the RV Y takes the value y .

- Note that $\mathbf{E}[X|Y = y]$ is a function of y .
- **Stick example:** stick of length l , break at uniformly chosen point Y , break again at uniformly chosen point X .
 - $\mathbf{E}[X|Y = y] = \frac{y}{2}$ (number)
 - $\mathbf{E}[X|Y] = \frac{Y}{2}$ (RV)

Law of iterated expectations

$$\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X|Y]] = \begin{cases} \sum_y \mathbf{E}[X|Y = y] p_Y(y), & Y \text{ discrete,} \\ \int_{-\infty}^{\infty} \mathbf{E}[X|Y = y] f_Y(y) dy, & Y \text{ continuous.} \end{cases}$$

- **Facilitate the calculation of expected values** when the problem data include conditional probabilities.
- In the stick example:

$$\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X|Y]] = \mathbf{E}[Y/2] = l/4.$$

- A class has n students and the quiz score of student i is x_i . The average quiz score is

$$m = \frac{1}{n} \sum_{i=1}^n x_i.$$

- The students are divided into k disjoint subsets $A_1, \dots, A_k$. We use n_s to denote the number of students in section s . The average score is

$$m_s = \frac{1}{n_s} \sum_{i \in A_s} x_i.$$

- Consider an experiment in which a student is selected with probability $1/n$. Define

X = quiz score of a student,

Y = section of a student, $Y \in \{1, \dots, k\}$.

- We then have

$$\mathbf{E}[X] = m.$$

- Conditioning on $Y = s$, every student in that section has the same probability $1/n_s$,

$$\mathbf{E}[X \mid Y = s] = \frac{1}{n_s} \sum_{i \in A_s} x_i = m_s.$$

- A randomly selected student belongs to section s with probability n_s/n , i.e., $\mathbf{P}(Y = s) = n_s/n$. Hence,

$$m = \mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X \mid Y]] = \sum_{s=1}^k \mathbf{E}[X \mid Y = s] \mathbf{P}(Y = s) = \sum_{s=1}^k \frac{n_s}{n} m_s.$$

The Conditional Expectation as an Estimator

Denote the conditional expectation

$$\hat{X} = \mathbf{E}[X | Y]$$

as an **estimator** of X given Y , and the estimation error

$$\tilde{X} = X - \hat{X}$$

is a RV.

- Properties of the estimator: **Unbiased**

- Conditional expectation of $\tilde{X}$ (for any possible $Y = y$)

$$\mathbf{E}[\tilde{X} | Y] = \mathbf{E}[X - \hat{X} | Y] = \mathbf{E}[X | Y] - \mathbf{E}[\hat{X} | Y] = \hat{X} - \hat{X} = 0.$$

- By the law of iterated expectations

$$\mathbf{E}[\tilde{X}] = \mathbf{E}[\mathbf{E}[\tilde{X} | Y]] = 0.$$

- Properties of the estimator: **Uncorrelated**

- $\hat{X}$ is uncorrelated with the estimation error $\tilde{X}$.

$$\mathbf{E}[\hat{X}\tilde{X}] = \mathbf{E}[\mathbf{E}[\hat{X}\tilde{X} | Y]] = \mathbf{E}[\hat{X}\mathbf{E}[\tilde{X} | Y]] = 0.$$

- The second equality follows from the fact that $\hat{X}$ is completely determined by Y .
- It follows that

$$\text{cov}(\hat{X}, \tilde{X}) = \mathbf{E}[\hat{X}\tilde{X}] - \mathbf{E}[\hat{X}]\mathbf{E}[\tilde{X}] = 0 - \mathbf{E}[X] \cdot 0 = 0.$$

- Since $X = \hat{X} + \tilde{X}$, the variance of X can be decomposed as

$$\text{var}(X) = \text{var}(\hat{X}) + \text{var}(\tilde{X})$$

- The first item $\text{var}(\hat{X})$ can be written as

$$\text{var}(\hat{X}) = \text{var}(\mathbf{E}[X | Y]).$$

- Consider the RV

$$\text{var}(X | Y) = \mathbf{E}[(X - \mathbf{E}[X | Y])^2 | Y] = \mathbf{E}[\tilde{X}^2 | Y]$$

and since $\mathbf{E}[\tilde{X}] = 0$

$$\text{var}(\tilde{X}) = \mathbf{E}[\tilde{X}^2] = \mathbf{E}[\mathbf{E}[\tilde{X}^2 | Y]] = \mathbf{E}[\text{var}(X | Y)].$$

Law of total variance

$$\text{var}(X) = \text{var}(\mathbf{E}[X | Y]) + \mathbf{E}[\text{var}(X | Y)]$$

- Two sections: $Y = 1$ (10 students); $Y = 2$ (20 students)
- We have

$$Y = 1 : \frac{1}{10} \sum_{i=1}^{10} x_i = 90. \quad Y = 2 : \frac{1}{20} \sum_{i=11}^{30} x_i = 60.$$

- Average scores

$$\mathbf{E}[X] = \frac{1}{30} \sum_{i=1}^{30} x_i = \frac{90 \cdot 10 + 60 \cdot 20}{30} = 70.$$

- Section average scores

$$\mathbf{E}[X \mid Y = 1] = 90, \quad \mathbf{E}[X \mid Y = 2] = 60.$$

- Using the law of iterated expectations,

$$\mathbf{E}[\mathbf{E}[X \mid Y]] = \frac{1}{3} \cdot 90 + \frac{2}{3} \cdot 60 = 70 = \mathbf{E}[X].$$

- Furthermore,

$$\text{var}(\mathbf{E}[X \mid Y]) = \frac{1}{3}(90 - 70)^2 + \frac{2}{3}(60 - 70)^2 = 200.$$

- Conditional variance

$$\text{var}(X \mid Y = 1) = \frac{1}{10} \sum_{i=1}^{10} (x_i - 90)^2 = 10.$$

$$\text{var}(X \mid Y = 2) = \frac{1}{20} \sum_{i=11}^{30} (x_i - 60)^2 = 20.$$

- Hence,

$$\mathbf{E}[\text{var}(X \mid Y)] = \frac{1}{3} \cdot 10 + \frac{2}{3} \cdot 20 = \frac{50}{3}.$$

- Therefore,

$$\text{var}(X) = \mathbf{E}[\text{var}(X \mid Y)] + \text{var}(\mathbf{E}[X \mid Y]) = \frac{50}{3} + 200.$$

- Average variability **within** sections + variability **between** sections

- We consider n independent tosses of a biased coin whose probability of heads, Y , is uniformly distributed over the interval $[0, 1]$.
- X is the number of heads. Calculate $\text{var}(X)$.

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Solution:

- We have $\mathbf{E}[X | Y] = nY$ and $\text{var}(X | Y) = nY(1 - Y)$. Thus,

$$\mathbf{E}[\text{var}(X | Y)] = \mathbf{E}[nY(1 - Y)] = n(\mathbf{E}[Y] - \mathbf{E}[Y^2]) = \frac{n}{6}.$$

$$\text{var}(\mathbf{E}[X | Y]) = \text{var}(nY) = \frac{n^2}{12}.$$

- Therefore,

$$\text{var}(X) = \mathbf{E}[\text{var}(X | Y)] + \text{var}(\mathbf{E}[X | Y]) = \frac{n}{6} + \frac{n^2}{12}.$$

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Definition of the transform (or moment generating function)

The transform associated with a RV X is a function $M_X(s)$ of a scalar parameter s , defined by

$$M_X(s) = \mathbf{E}[e^{sX}] = \begin{cases} \sum_x p_X(x)e^{sx}, & \text{discrete RV} \\ \int_{-\infty}^{\infty} f_X(x)e^{sx} dx, & \text{continuous RV} \end{cases}$$

• Remarks

- a function of s , rather than a number
- not necessarily defined for all (complex) s
- always well defined for $\text{Re}(s) = 0$
- compared with Laplace transform

- Sanity checks

$$M_X(0) = 1$$

$$|M_X(s)| \leq 1 \text{ for } \operatorname{Re}(s) = 0$$

- Linear operations

$$M_{aX}(s) = M_X(as)$$

$$M_{X+c}(s) = M_X(s)e^{cs}$$

$$M_{X+Y}(s) = M_X(s)M_Y(s) \text{ if } X, Y \text{ independent}$$

- Expected values

$$\mathbf{E}[X^n] = \left. \frac{\partial^n M_X(s)}{\partial s^n} \right|_{s=0}$$

$$\mathbf{P}(X = c) = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N M_X(jk) e^{-jk c}$$

- Let X be a Poisson RV with parameter λ

$$p_X(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \dots$$

- The corresponding transform

$$M(s) = \sum_{x=0}^{\infty} e^{sx} \frac{\lambda^x e^{-\lambda}}{x!} = e^{\lambda(e^s - 1)}.$$

- Let X be an exponential RV with parameter λ

$$f_X(x) = \lambda e^{-\lambda x}, \quad x \geq 0.$$

- The corresponding transform

$$M(s) = \lambda \int_{x=0}^{\infty} e^{sx} e^{-\lambda x} dx = \frac{\lambda}{\lambda - s}.$$

- Let $M_X(s)$ be the transform associated with a RV X . Consider $Y = aX + b$. We then have

$$M_Y(s) = \mathbf{E}[e^{s(aX+b)}] = e^{sb}\mathbf{E}[e^{saX}] = e^{sb}M_X(sa).$$

- Let $M_X(s)$ be the transform associated with a RV X . Consider $Y = aX + b$. We then have

$$M_Y(s) = \mathbf{E}[e^{s(aX+b)}] = e^{sb}\mathbf{E}[e^{saX}] = e^{sb}M_X(sa).$$

- Let Y be a standard normal RV, the associated transform is

$$M_Y(s) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(y^2/2)} e^{sy} dy = e^{s^2/2}.$$

- Consider $X = \sigma Y + \mu$, we obtain

$$M_X(s) = e^{s\mu} M_Y(s\sigma) = e^{(\sigma^2 s^2/2) + \mu s}.$$

- Consider a continuous RV X , take the derivative

$$\frac{d}{ds}M(s) = \frac{d}{ds} \int_{-\infty}^{\infty} e^{sx} f_X(x) dx = \int_{-\infty}^{\infty} x e^{sx} f_X(x) dx.$$

- For the special case where $s = 0$,

$$\left. \frac{d}{ds}M(s) \right|_{s=0} = \int_{-\infty}^{\infty} x f_X(x) dx = \mathbf{E}[X].$$

- More generally, if we differentiate n times the function $M(s)$,

$$\left. \frac{d^n}{ds^n}M(s) \right|_{s=0} = \int_{-\infty}^{\infty} x^n f_X(x) dx = \mathbf{E}[X^n].$$

Moment generating properties

$$M_X(0) = 1, \quad \left. \frac{d}{ds}M_X(s) \right|_{s=0} = \mathbf{E}[X], \quad \left. \frac{d^n}{ds^n}M_X(s) \right|_{s=0} = \mathbf{E}[X^n]$$

- For an exponential RV X with parameter λ , the associated transform is

$$M(s) = \frac{\lambda}{\lambda - s}.$$

- Thus,

$$\frac{d}{ds}M(s) = \frac{\lambda}{(\lambda - s)^2}, \quad \frac{d^2}{ds^2}M(s) = \frac{2\lambda}{(\lambda - s)^3}.$$

- By setting $s = 0$, we obtain

$$\mathbf{E}[X] = \frac{1}{\lambda}, \quad \mathbf{E}[X^2] = \frac{2}{\lambda^2}.$$

Inversion Property

The transform $M_X(s)$ associated with a RV X uniquely determines the CDF of X , assuming that $M_X(s)$ is finite for all s in some interval $[-a, a]$, where a is a positive number.

Example:

- Given the transform associated with a RV X is of the form

$$M(s) = \frac{pe^s}{1 - (1-p)e^s},$$

where p is a constant and $0 < p \leq 1$.

- Recall the formula for the geometric series,

$$\frac{1}{1-\alpha} = 1 + \alpha + \alpha^2 + \cdots,$$

- We use this formula with $\alpha = (1 - p)e^s < 1$ to obtain

$$\begin{aligned} M(s) &= pe^s(1 + (1 - p)e^s + (1 - p)^2e^{2s} + \cdots) \\ &= \sum_{k=1}^{\infty} p(1 - p)^{k-1}e^{ks} \end{aligned}$$

- The probability $\mathbf{P}(X = k)$ is found by reading the coefficient of the term e^{ks} .
- Therefore,

$$\mathbf{P}(X = k) = p(1 - p)^{k-1}, \quad k = 1, 2, \dots$$

- Let $X_1, \dots, X_n$ be continuous RVs with PDFs $f_{X_1}, \dots, f_{X_n}$.
- The value y of RV Y is generated as follows: an index i is chosen with a corresponding probability p_i , and y is taken to be equal to the value of X_i . Then,

$$f_Y(y) = p_1 f_{X_1}(y) + \dots + p_n f_{X_n}(y)$$

- With transform function

$$M_Y(s) = p_1 M_{X_1}(s) + \dots + p_n M_{X_n}(s)$$

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1. Let X , Y and Z be discrete RVs. Show the following generalizations of the law of iterated expectation.

(a) $\mathbf{E}[Z] = \mathbf{E}[\mathbf{E}[Z \mid X, Y]]$.

(b) $\mathbf{E}[Z \mid X] = \mathbf{E}[\mathbf{E}[Z \mid X, Y] \mid X]$.

(c) $\mathbf{E}[Z] = \mathbf{E}[\mathbf{E}[\mathbf{E}[Z \mid X, Y] \mid X]]$.

2. Widgets are stored in boxes, and then all boxes are assembled in a crate. Let X be the number of widgets in any particular box, and N be the number of boxes in a crate. Assume that X and N are independent integer-valued RVs, with expected value equal to 10, and variance equal to 16. Evaluate the expected value and variance of T , where T is the total number of widgets in a crate.

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- Conditional expected value

$$\mathbf{E}[X|Y = y] = \sum_x xp_{X|Y}(x|y)$$
$$\mathbf{E}[\mathbf{E}[X|Y]] = \sum_y \mathbf{E}[X|Y = y]p_Y(y) = \mathbf{E}[X]$$

- Conditional variance

$$\text{var}(X | Y) = \mathbf{E}[(X - \mathbf{E}[X | Y])^2 | Y] = \mathbf{E}[\tilde{X}^2 | Y]$$

$$\text{var}(X) = \mathbf{E}[\text{var}(X | Y)] + \text{var}(\mathbf{E}[X | Y])$$

- Transforms

$$M_X(s) = \mathbf{E}[e^{sX}]$$

- Readings: Section 4.3-4.4
- Assignment: Problem Set 10
- Next lecture: Sum of a random number of RVs